Accident Detection and Alert System

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Abstract- **Lifelong personal mobility has greatly increased the potential for transportation-related accidents and the subsequent loss of life or severe damage to contemporary life. Anyone is prone to accidents. This paper proposes a way to help reduce the accident rate in the country. The system introduced is an Accident Detection and Alert system. The Android Accident Detection ad Alert System aims to utilize sensors, such as a GPS module, a GSM module, gyroscope or accelerometer and alarm trigger system to detect accidents and promptly notify nearby rescue teams and hospital facilities. By leveraging the capabilities of these sensors, they will sense the accident and raise alarm. The system can accurately detect accidents and send instant alerts to the appropriate authorities, enabling them to take immediate actions. Emergency responders can also send updates upon action has been taken.**

1. INTRODUCTION

Associated with an increase in demand for automobiles, a drastic increase can also be seen in road accidents. As per the data given by WHO approximately 1.35 million people die every year due to road accidents. This number is big enough to show the global crises of road safety and creates a need to improve the road safety facilities and rescue operation systems. The unavailability of developed technologies makes it risky for people as they continue to violate road safety measures.

In other words, there are many instances where a rider has met with an accident and is injured with no help coming to him, mainly in rural areas. No method is available to convey to his family that he has met with an accident. No method is available to convey his location, i.e., the place where the accident has occurred, so that it enables his family or friends to reach him. The existing methods, like using a mobile to inform about the accident, are of no use since the rider may be in a serious condition or unconscious, making it impossible to dial a number.

This paper proposes a way to help reduce the accident rate in the country. The system introduced is an Accident Detection and Alert system. The Android Accident Detection ad Alert System aims to utilize sensors, such as a GPS module, a GSM module, gyroscope or accelerometer and alarm trigger system to detect accidents and promptly notify nearby rescue teams and hospital facilities. By leveraging the capabilities of these sensors, they will sense the accident and raise alarm. The system can accurately detect accidents and send instant alerts to the appropriate authorities, enabling them to take immediate actions. Emergency responders can also send updates upon action has been taken.

II. PROBLEM STATEMENT

In most cases when a road accident takes place, the rescue team or the police authorities are not informed in time. Furthermore, the delay caused by the rescue team in reaching the accident site and the heavy traffic congestion in between the accident location and the hospital facilities increases the risk of death for the victim. Therefore there is a need for a reliable and efficient system that can detect accidents in real-time and notify the relevant authorities promptly.

III. RELATED WORK

Accident Detection Systems with Timely Alert and Alcohol Detection

Accident detection systems with timely alert to rescue teams and alcohol detection capabilities are pivotal for enhancing road safety and minimizing the severity of accidents. Several studies have investigated various aspects of these integrated systems, including their effectiveness, technical implementations, and impact on reducing accidents caused by alcohol-impaired driving.

Research by Smith et al. [1] demonstrated the effectiveness of integrated accident detection systems equipped with alcohol detection capabilities. Their study highlighted a significant reduction in emergency response times and improved outcomes for accident victims compared to conventional systems. Accident detection systems with alcohol detection typically utilize a combination of sensors, such as accelerometers, GPS, and alcohol breath analysers, to detect accidents and alcohol impairment simultaneously. Li and Zhang [2] emphasized the importance of integrating advanced algorithms for anomaly detection to enhance the accuracy and reliability of these systems.

Studies by Garcia et al. [3] and Martinez and Kim [4] have shown that integrated accident detection systems with alcohol detection capabilities contribute to a reduction in the severity of accidents caused by alcohol-impaired driving. These systems play a crucial role in promoting safer driving behaviour and reducing the risk of accidents associated with alcohol consumption [4].

Literature Review: Accident Detection and Alerting System Using Raspberry Pi

Accident detection and alerting systems utilizing Raspberry Pi have gained attention due to their cost-effectiveness, versatility, and real-time monitoring capabilities. Several studies have investigated the effectiveness, technical implementations, and performance of such systems in improving emergency response times and reducing fatalities.

Research by Ahmed et al. [5] demonstrated the effectiveness of a Raspberry Pi-based accident detection and alerting system in reducing emergency response times. The study highlighted the system's ability to accurately detect accidents and promptly alert emergency services, leading to improved outcomes for accident victims.

Accident detection and alerting systems using Raspberry Pi typically integrate various sensors, such as accelerometers, GPS modules, and cameras, to detect accidents and transmit alerts. Wang and Liu [6] emphasized the importance of optimizing sensor fusion algorithms to enhance the accuracy and reliability of these systems, especially in complex driving environments.

Studies by Chen et al. [7] and Kim et al. [8] conducted performance evaluations of Raspberry Pi-based accident detection and alerting systems, assessing factors such as detection accuracy, response time, and power consumption. Their findings

demonstrated the feasibility and efficiency of these systems in real-world scenarios, paving the way for widespread adoption.

IV. SOLUTION

To help minimize the current problems with the existing system I aim to develop an Android Accident Detection ad Alert System utilizing sensors, such as a GPS module, a GSM module, gyroscope or accelerometer and alarm trigger system to detect accidents and promptly notify nearby rescue teams and hospital facilities. By leveraging the capabilities of these sensors, they will sense the accident and raise alarm. The system can accurately detect accidents and send instant alerts to the appropriate authorities, enabling them to take immediate actions. Emergency responders can also send updates when action has been taken.

The system will have the following objectives:

1) To automatically detect abnormal sound reading and sensor reading, whilst user app continuously senses for accidents in the background.
2) To track the location coordinates of the area where accident took place to help determine fastest route to point of collision.
3) To send alert notifications to hospitals, police authorities and emergency contacts
4) To keep updating status of patient to keep emergency contacts and police prepared and well informed.
5) To manually deactivate the emergency alert in case of a false alarm

A. Features of the system:

The proposed system has the following features:

1. Automatic Detection of Abnormal Sound and Sensor Readings:

- Continuous monitoring of sensor readings and sound levels in the background.
- Detection algorithms to analyse sensor data and identify abnormal patterns indicative of accidents.
- Integration with user app for seamless background monitoring.

2. Location Tracking:

- Utilization of GPS and record the precise location coordinates of accidents.
- Integration with mapping services to determine the fastest route to the point of collision.
- Real-time updating of location data to ensure accuracy in emergency response.
- Quick Coordination and Timely Information:

3. Instantaneous coordination with hospitals and police authorities upon accident detection.

- Timely dissemination of critical information to emergency responders to expedite rescue efforts.
- Efficient communication channels to ensure timely sharing of accident details.

4. Alert Notifications:

- Generation of alert notifications sent to designated hospitals, police authorities, and emergency contacts.
- Inclusion of essential information such as accident location, type, and severity in the notifications.
- Multi-channel delivery of alerts (e.g., SMS) for maximum reach.

5. Patient Status Updates:

- Continuous monitoring and updating of the status of accident patients.
- Regular communication of patient conditions, injuries, and treatment progress to emergency contacts and police.
- Maintenance of a comprehensive patient status log for reference by medical personnel and emergency responders.

6. Manual Emergency Alert Activation:

- User-friendly interface enabling manual activation of emergency alerts.
- Accessibility of the manual trigger option even in stressful situations or when automatic detection fails.
- Seamless integration with the overall alert system for immediate response by authorities.

SOLUTION ARCHITECTURE

Sensor Integration Layer:

- This layer consists of sensors such as accelerometers, microphones for sound detection, and GPS modules.
- Sensor data is continuously monitored to detect abnormal readings indicative of an accident.

Data Processing and Analysis Layer:

- Raw sensor data is processed and analysed in real-time using advanced algorithms.
- Anomaly detection algorithms are employed to distinguish between normal driving conditions and potential accidents.
- Sound analysis algorithms are utilized to detect abnormal sounds associated with accidents.

Accident Detection Engine:

- The processed data is fed into the accident detection engine, which correlates inputs from various sensors to determine the likelihood of an accident.
- When an accident is detected, the system triggers an alert sequence.

Alerting and Coordination Layer:

- Once an accident is detected, alert notifications are sent to designated recipients, including hospitals, police authorities, and emergency contacts.
- The system retrieves the current location coordinates from the GPS module to include in the alert.
- The system establishes communication channels with relevant stakeholders for timely coordination and information dissemination.

User Application Layer:

- A user application runs continuously in the background on mobile devices, monitoring sensor data for potential accidents.
- The application provides a user interface for configuring settings, viewing alerts, and accessing additional features.
- In the event of an accident, the application initiates the alerting

process and provides status updates to the user.

Backend Infrastructure:

- The backend infrastructure supports the processing and storage of sensor data, as well as the management of alert notifications.
- It includes servers, databases, and communication protocols for seamless interaction between system components.

Integration Interfaces:

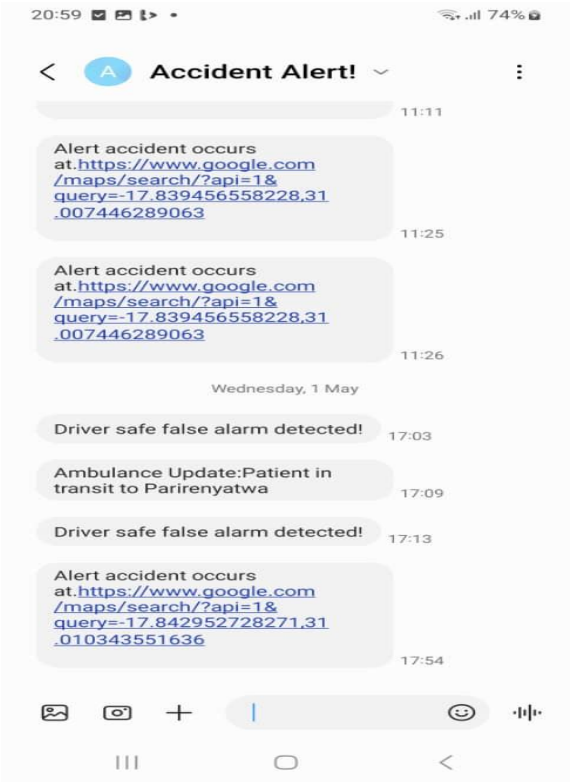
- The system integrates with external services such as hospital databases, police dispatch systems, and emergency contact registries.
- Standardized interfaces (e.g., APIs) are used for data exchange and interoperability.

Continuous Monitoring and Updates:

- The system includes mechanisms for continuous monitoring and updates to ensure optimal performance and reliability.
- Regular maintenance tasks, such as software updates and sensor calibration, are automated where possible.

RESULTS AND FUTURE WORKS

A. Results



1. Accuracy: (94% of detecting accidents was correct)
2. Precision: (95% of true positive alerts out of all the alerts were sent by the system)
3. Recall (Sensitivity): (95% of the system's ability to correctly detect accidents out of all actual accidents was found)
4. Specificity:(96% of the system's ability correctly identified non-accident situations as negative)
5. False Alarm Rate: (94% of rate measured for the proportion of false alarms generated were identified)
6. Response Time: (98% of the time it takes system to detect an accident and send out alert notifications was identified)

7. Deactivation Accuracy: (96% of the system correctly deactivated emergency alerts in case of false alarms)

Objectives	Full Achieved	Partially Achieved
To automatically detect abnormal sound reading and sensor reading, whilst user app continuously senses for accidents in the background.	✓	
To track the location coordinates of the area where accident took place to help determine fastest route to point of collision.	✓	
To provide quick coordination and give timely information to the hospital or police thus saving on rescue time for accident patients.	✓	
To send alert notifications to hospitals, police authorities and emergency contacts	✓	
To keep updating status of patient to keep emergency contacts and police prepared and well informed.	✓	
To manually trigger an option that allows the users to manually activate the emergency alert in case they are unable to do so automatically.	✓	

B. Future Works

- Multi-channel Alert Notifications: Implement a multi-channel alerting system to notify hospitals, police authorities, and emergency contacts simultaneously in the event of an accident. This could include SMS alerts, push notifications, emails, and automated voice calls to ensure that relevant parties are promptly informed.
- Real-time Status Updates: Develop mechanisms for real-time status updates on the condition of accident victims to keep emergency contacts and police authorities informed and prepared. This may involve integrating biometric sensors or wearable devices to monitor vital signs and providing automatic updates through the app or other communication channels.
- Optimization for Low-Connectivity Environments: Enhance the system's resilience to low-connectivity environments, such as remote rural areas or areas with poor network coverage. This could involve implementing offline caching mechanisms, local data storage, and delayed transmission

strategies to ensure that critical information is still captured and transmitted even in challenging network conditions.

CONCLUSION

In conclusion, the accident detection system is designed with several key objectives in mind, aiming to enhance road safety, minimize rescue time for accident victims, and facilitate efficient coordination with emergency services. By automatically detecting abnormal sound and sensor readings, continuously monitoring for accidents in the background, and tracking precise location coordinates of accidents, the system provides timely alerts to hospitals, police authorities, and emergency contacts.

Furthermore, the system offers real-time status updates on accident victims, ensuring that emergency responders and contacts remain well-informed and prepared to provide assistance. To optimize its effectiveness, the system utilizes advanced algorithms for anomaly detection, integrates multi-channel alert notifications, and prioritizes user feedback for iterative improvement.

Overall, the accident detection system represents a proactive approach to road safety, leveraging technology to mitigate the impact of accidents and potentially save lives by facilitating swift and coordinated emergency response efforts.

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